AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	VA facility VA_way_1333205263 -- station KJYO, America/New_York
Committed pair	OSM 1128589933 -> 1333205263 OSM way 1128589933 / OSM way 1333205263
Facade gap	189.4 m between the two halls
Weather record	43,249 real hourly records from KJYO
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 1.0172 C at 355 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2021-03-25 (clear_cool)
Plant limit	18.0 C
Notice required	6 h
Forecast skill	0.00 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 10 of 23 hours with 2 mode changes. 7 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	10 of 23 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent	10 of 23 hours free cooling, 2 mode change(s), 0 unsafe hour(s) 7 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	8.235	18.0	9.000	3.993
01	FREE	yes	-	8.235	18.0	9.000	3.340
02	FREE	yes	-	9.235	18.0	9.000	3.688
03	FREE	yes	-	9.235	18.0	9.000	3.191
04	FREE	yes	-	9.235	18.0	10.000	2.855
05	FREE	yes	-	9.235	18.0	9.000	2.477
06	FREE	yes	-	9.091	18.0	10.000	1.915
07	FREE	yes	-	11.123	18.0	10.000	2.749

08	FREE	yes	-	13.088	18.0	10.000	2.730
09	FREE	yes	-	16.087	18.0	10.000	3.788
10	mechanical	no	dry-bulb	20.080	18.0	10.000	5.092
11	mechanical	no	dry-bulb	21.073	18.0	11.000	5.685
12	mechanical	no	dry-bulb	23.073	18.0	11.000	5.731
13	mechanical	no	dry-bulb	23.077	18.0	11.000	5.806
14	mechanical	no	dry-bulb	21.235	18.0	12.000	5.100
15	mechanical	no	refusal	19.172	18.0	13.000	4.599
16	mechanical	yes	switch budget	16.105	18.0	14.000	3.268
17	mechanical	yes	switch budget	15.056	18.0	14.000	3.058
18	mechanical	yes	switch budget	13.049	18.0	14.000	2.841
19	mechanical	yes	switch budget	12.235	18.0	13.000	3.789
20	mechanical	yes	switch budget	11.082	18.0	13.000	3.085
21	mechanical	yes	switch budget	12.082	18.0	12.000	3.973
23	mechanical	yes	switch budget	13.073	18.0	13.000	4.271

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.235 C, which is 9.765 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.993 C, and the margin is measured, not chosen: 3.757 C of group-conditional forecast error for this hour of day, plus 0.2352 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 8.235 C, which is 9.765 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.340 C, and the margin is measured, not chosen: 3.105 C of group-conditional forecast error for this hour of day, plus 0.2352 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.235 C, which is 8.765 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.688 C, and the margin is measured, not chosen: 3.453 C of group-conditional forecast error for this hour of day, plus 0.2352 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.235 C, which is 8.765 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.191 C, and the margin is measured, not chosen: 2.956 C of group-conditional forecast error for this hour of day, plus 0.2352 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.235 C, which is 8.765 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.855 C, and the margin is measured, not chosen: 2.620 C of group-conditional forecast error for this hour of day, plus 0.2352 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.235 C, which is 8.765 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.477 C, and the margin is measured, not chosen: 2.241 C of group-conditional forecast error for this hour of day, plus 0.2352 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.091 C, which is 8.909 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.915 C, and the margin is measured, not chosen: 1.824 C of group-conditional forecast error for this hour of day, plus 0.0912 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.123 C, which is 6.877 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.749 C, and the margin is measured, not chosen: 2.626 C of group-conditional forecast error for this hour of day, plus 0.1230 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

08:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 13.088 C, which is 4.912 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.730 C, and the margin is measured, not chosen: 2.642 C of group-conditional forecast error for this hour of day, plus 0.0877 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

09:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.087 C, which is 1.913 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 3.788 C, and the margin is measured, not chosen: 3.702 C of group-conditional forecast error for this hour of day, plus 0.0868 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.080 C against a 18.0 C limit, so it fails by 2.080 C. It would take a limit 2.080 C higher, or a bound 2.080 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.073 C against a 18.0 C limit, so it fails by 3.073 C. It would take a limit 3.073 C higher, or a bound 3.073 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.073 C against a 18.0 C limit, so it fails by 5.073 C. It would take a limit 5.073 C higher, or a bound 5.073 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.077 C against a 18.0 C limit, so it fails by 5.077 C. It would take a limit 5.077 C higher, or a bound 5.077 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.235 C against a 18.0 C limit, so it fails by 3.235 C. It would take a limit 3.235 C higher, or a bound 3.235 C tighter, to change this hour.

15:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

16:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.105 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

17:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.056 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.049 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

19:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.235 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

20:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.081 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

21:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 12.081 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 13.073 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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